

# README

## Architecture — Model-Access System (Index)

| Field         | Value                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Revision      | 1                                                                                                                                 |
| Created       | 2026-06-10                                                                                                                        |
| Last modified | 2026-06-10                                                                                                                        |
| Status        | DESIGN index — IMPLEMENTED vs PLANNED clearly separated in each doc                                                               |
| Maintainer    | DOCUMENTATION subagent (READ-ONLY on code)                                                                                        |
| Authority     | CONST-036 (LLMsVerifier single source of truth), CONST-040 (capabilities from <code>VerificationResult</code> ), §11.4 anti-bluff |

This folder documents the **model-access + provider-exposure + CLI-bridge** system: how a recognized API key turns into the WORKING models a user can use, how every provider/ensemble/ HelixLLM/CLI-agent is exposed under one uniformly-named root, and how the CLI-bridge proxies prompts through real CLI binaries.

### Documents (this index)

| File                                         | What it covers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Primary sources                                      |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| <a href="#">model-access-architecture.md</a> | Mermaid diagrams: <b>(A)</b> key-recognition → verifier → working-model funnel (SP1); <b>(B)</b> unified catalog joining ensemble + HelixLLM + each provider + each verified model + each CLI-agent under one root (SP2/SP4) with the <code>ensemble/helixllm/&lt;provider&gt;/&lt;model&gt;/cli/&lt;agent&gt;/&lt;model&gt;</code> naming; <b>(C)</b> the CLI-bridge selection + proxy sequence (PRIMARY system-installed → FALLBACK submodule-built → proxy → parse → power-features). Includes a node-by-node IMPLEMENTED-vs-PLANNED provenance ledger. | SP1/SP2/SP4 plans; analyses B/C/D; master roadmap    |
| <a href="#">model-access-schema.sql</a>      | DESIGN SQL DDL for the data model: providers, models,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <code>helix_code/internal/verifier/types.go</code> ; |

| File | What it covers                                                                                                                                                                                                                                                   | Primary sources                                            |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
|      | verification_results, capabilities, catalog_entries, cli_agent_bridges, api_key_sources + a working_models view. Each column tagged SOT=verifier (CONST-036/040 verifier-owned) or SOT=local. Grounded in the real VerifiedModel/ VerificationResult Go structs. | submodules/<br>llms_verifier/.../<br>llmverifier/models.go |

Other files already present in this directory (CONST-051-tension.md, CONST046-i18n-overview.md) predate this index and cover unrelated concerns.

## Honesty note (§11.4 / §11.4.123)

Both documents describe the **target architecture**. They explicitly separate **IMPLEMENTED** components (real code, cited file:line) from **PLANNED** components (specified in SP1/SP2/SP4 but not yet built — no captured runtime evidence). A PLANNED node is a design intent, not a claim of working software. The runtime-evidence bar (§11.4.5 / §11.4.69 / §11.4.107) is met only by the tests/Challenges each SP plan requires at execution time — not by these diagrams or this DDL.

Quick orientation on current state (from the analysis passes):

- **IMPLEMENTED today:** real LLMsVerifier REST consumption (verifier/client.go, adapter.go:183-224); viper key binding (config.go:307-345); the CLI-exec primitive (base.go:120-138); config re-export + static Validate() (unified\_cli\_generator.go).
- **DEAD/UNWIRED (wire, don't delete — §11.4.124):** secrets.LoadAPIKeys (loader.go:30).
- **Known bluffs to remediate:** CLI shows failed/pending models (D-2), /v1/completion/models hardcoded 3-model list (D-1), hardcoded provider lists (D-5/CONST-036), qwencode/instance\_manager CLI stubs (D-6).
- **PLANNED (not built):** the GetWorkingModels filter, key-presence gate, unified GET /v1/catalog, HelixLLM root promotion, the CLI AgentProvider adapter, replace dev.helix.agent wiring (D-7), filesystem INSTALL + LIVE post-install validation.

## Upstream references

- SP plans: docs/superpowers/specs/plans/2026-06-10-SP{1,2,4}-\*.md
- Analyses: docs/superpowers/specs/analysis/2026-06-10-{B,C,D}-\*.md
- Master roadmap: docs/superpowers/specs/2026-06-10-llms-access-master-roadmap.md

Per CONST-066 / §11.4.65, the .md documents should ship synchronized .html/.pdf siblings at release-prep. Those exports are NOT generated by this READ-ONLY documentation pass.